

3. EXCLUSIVE COMMUNICATIONS

The Exclusive Messages can be transmitted or recognized only when the Exclusive switch on the S-550 is ON, Ignored when OFF.

The Model-ID number of the S-550 is [1EH].

Device-ID can be changed from the panel in MIDI Mode.
The numbers 1-16 on the display correspond to Device-ID codes 0-15, respectively.

Each Address and Size should be 4 bytes of data, respectively.

3.1 One way communication

3.1.1 Request RQ1 11H

Only when the recognized address and size in RQ1 match those on the S-550, it transmits the corresponding data.

It ignores Requests having illegal address or size. *3-1

The S-550 won't transmit RQ1.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
11H	Command-ID (RQ1)	
aaH	Address MSB	*3-1
aaH	Address	
aaH	Address	
aaH	Address LSB	
ssH	Size MSB	*3-1
ssH	Size	
ssH	Size	
ssH	Size LSB	
sum	Checksum	
F7H	EOX (End of Exclusive)	

3.1.2 Data set DT1 12H

When the recognized Dataset message contains an appropriate address and size data, the S-550 stores the associated data that address.
It ignores any Data set having illegal address.

The S-550 transmits a Data set message when a Tone Parameter is edited on the pannel or when the S-550 recognizes RQ1.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
12H	Command-ID (DT1)	
aaH	Address MSB	*3-1
aaH	Address	
aaH	Address	
aaH	Address LSB	
ddH	Data	*3-2
:		
sum	Checksum	
F7H	EOX (End of Exclusive)	

3.2 Handshaking communication

3.2.1 Want to send data WSD 40H

When recognized WSD message has an appropriate address and size data, the S-550 transmits ACK and waits the associated data.

If not appropriate, it will transmit RJC. *3-1

The S-550 won't transmit WSD.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
40H	Command-ID (WSD)	
aaH	Address MSB	*3-1
aaH	Address	
aaH	Address	
aaH	Address LSB	
ssH	Size MSB	*3-1
ssH	Size	
ssH	Size	
ssH	Size LSB	
sum	Checksum	
F7H	EOX (End of Exclusive)	

3.2.2 Request data RQD 41H

When recognized RQD message has an appropriate address and size data, the S-550 transmits the corresponding data.

If not appropriate, it will transmit RJC. *3-1

The S-550 won't transmit RQD.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
41H	Command-ID (RQD)	
aaH	Address MSB	*3-1
aaH	Address	
aaH	Address	
aaH	Address LSB	
ssH	Size MSB	*3-1
ssH	Size	
ssH	Size	
ssH	Size LSB	
sum	Checksum	
F7H	EOX (End of Exclusive)	

3.2.3 Data set DAT 42H

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
42H	Command-ID (DAT)	
aaH	Address MSB	*3-1
aaH	Address	
aaH	Address	
aaH	Address LSB	
ddH	Data	*3-2
:		
sum	Checksum	
F7H	EOX (End of Exclusive)	

3.2.4 Acknowledge ACK 43H

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
43H	Command-ID (ACK)	
F7H	EOX (End of Exclusive)	

3.2.5 End of data EOD 45H

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
45H	Command-ID (EOD)	
F7H	EOX (End of Exclusive)	

3.2.6 Communication error ERR 4EH

The S-550 transmits ERR if a checksum error occurs.

When ERR message is recognized, the S-550 transmits RJD and ceases the current communication.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
4EH	Command-ID (ERR)	
F7H	EOX (End of Exclusive)	

3.2.7 Rejection RJC 4FH

The S-550 transmits RJC and ceases communication if it detects one of the following:

- ERR is recognized,
- address in the recognized Dat set is not continuous one and
- ENTER is activated on the panel during communication.

Byte	Description	
F0H	Exclusive status	
41H	Roland-ID	
DEV	Device-ID	
1EH	Model-ID (S-550)	
4FH	Command-ID (RJC)	
F7H	EOX (End of Exclusive)	

Notes:

*3-1 Address and size should specify a memory space in which data exist.
The lowest bit of LSB byte in address and size should be 0.

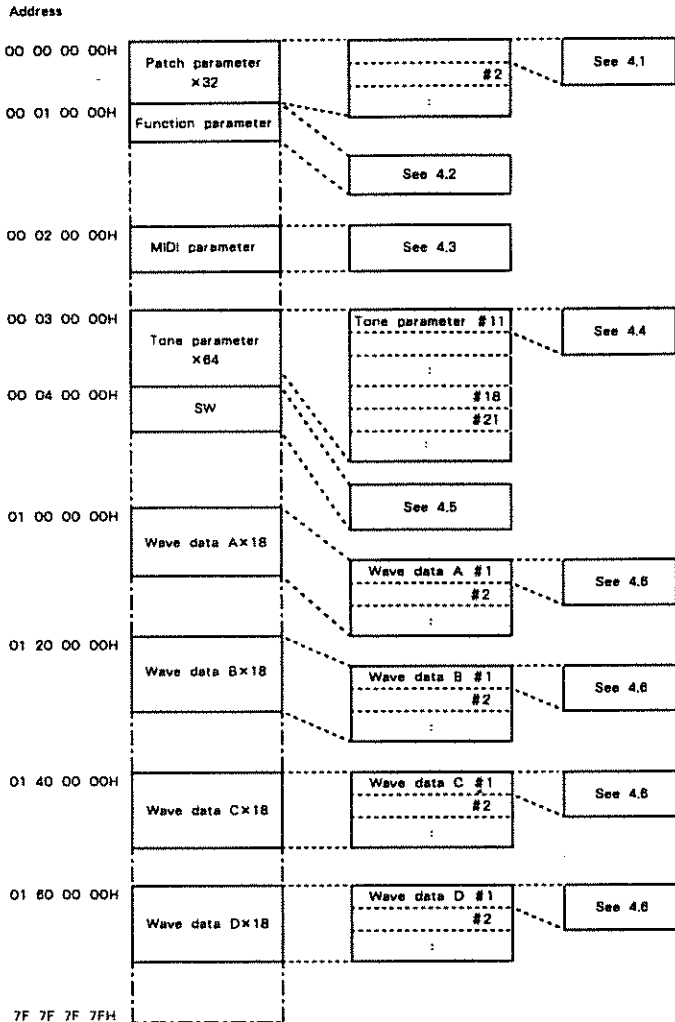
*3-2 The number of data bytes should be even number.

4. Address mapping of parameters

Address is represented from 00 to 7F by hexadecimal.

Address	MSB			LSB
binary	0aaa aaaa	0bbb bbbb	0ccc cccc	0ddd dddd
7 bit Hex	AA	BB	CC	DD

An offset address added to an address of each block makes a real address.



4.1 Patch parameter

Offset address	Description
00 00H	0000 aaaa PATCH NAME 1
00 01H	0000 bbbb aaaa bbbb 32-127 (ASCII)
:	
00 16H	0000 aaaa PATCH NAME 12
00 17H	0000 bbbb aaaa bbbb 32-127 (ASCII)
:	
00 18H	0000 aaaa BEND RANGE
00 19H	0000 bbbb aaaa bbbb 0-12
:	
00 1AH	0xxx xxxx dummy
00 1BH	0xxx xxxx
:	
00 1CH	0000 aaaa AFTER TOUCH SENSE
00 1DH	0000 bbbb aaaa bbbb 0-127
:	
00 1EH	0000 aaaa KEY MODE
00 1FH	0000 bbbb aaaa bbbb 0: Normal 1: V-Sw 2: X-Fade 3: V-Mix 4: Unison
:	
00 20H	0000 aaaa VELOCITY SW THRESHOLD
00 21H	0000 bbbb aaaa bbbb 0-127
:	
00 22H	0000 aaaa TONE TO KEY #1-109
00 23H	0000 bbbb aaaa bbbb 0-31

:			-1: OFF
01 7AH	0000 aaaa	TONE TO KEY #1-109	
01 7BH	0000 bbbb	aaaa bbbb	0-31
:			
01 7CH	0000 aaaa	TONE TO KEY #2-1	
01 7DH	0000 bbbb	aaaa bbbb	0-31
:			
03 54H	0000 aaaa	TONE TO KEY #2-109	
03 55H	0000 bbbb	aaaa bbbb	0-31
:			
03 56H	0000 aaaa	COPY SOURCE	
03 57H	0000 bbbb	aaaa bbbb	0-7
:			
03 58H	0000 aaaa	OCTAVE SHIFT	
03 59H	0000 bbbb	aaaa bbbb	-2-+2
:			
03 5AH	0000 aaaa	OUTPUT LEVEL	
03 5BH	0000 bbbb	aaaa bbbb	0-127
:			
03 5CH	0000 aaaa	0xxx xxxx dummy	
03 5DH	0000 bbbb	0xxx xxxx	
:			
03 5EH	0000 aaaa	DETUNE	
03 5FH	0000 bbbb	aaaa bbbb	-64-+63
:			
03 60H	0000 aaaa	VELOCITY MIX RATIO	
03 61H	0000 bbbb	aaaa bbbb	0-127
:			
03 62H	0000 aaaa	AFTER TOUCH ASSIGN	
03 63H	0000 bbbb	aaaa bbbb	0: Modulation 1: Volume 2: Bend + 3: Bend - 4: Filter
:			
03 64H	0000 aaaa	KEY ASSIGN	
03 65H	0000 bbbb	aaaa bbbb	0: Rotary 1: Fix
:			
03 66H	0000 aaaa	OUTPUT ASSIGN	
03 67H	0000 bbbb	aaaa bbbb	0: OUTPUT 1 1: OUTPUT 2 2: OUTPUT 3 3: OUTPUT 4 4: OUTPUT 5 5: OUTPUT 6 6: OUTPUT 7 7: OUTPUT 8 8: TONE
:			
03 68H	0xxx xxxx	dummy	
:			
03 7FH	0xxx xxxx		
Total size	00 00 04 00H		

4.2 Function parameter

Offset address	Description
00 00H	0000 aaaa MASTER TUNE
00 01H	0000 bbbb aaaa bbbb -64-+63
:	
00 02H	0xxx xxxx dummy
:	
00 1BH	0xxx xxxx
:	
00 1CH	0xxx xxxx dummy
00 1DH	0xxx xxxx
:	
00 1EH	0xxx xxxx dummy
00 1FH	0xxx xxxx
:	
00 20H	0000 aaaa VOICE MODE
00 21H	0000 bbbb aaaa bbbb 0-23 0: AUTO MODE LAST NOTE PRIORITY 1: AUTO MODE FIRST NOTE PRIORITY 2-23: FIX MODE 1-22
:	
00 22H	0000 aaaa MULTI MIDI RX-CH 1
00 23H	0000 bbbb aaaa bbbb 0-15
:	
00 30H	0000 aaaa MULTI MIDI RX-CH 8
00 31H	0000 bbbb aaaa bbbb 0-15
:	
00 32H	0000 aaaa MULTI PATCH NUMBER 1
00 33H	0000 bbbb aaaa bbbb 0-31
:	
00 40H	0000 aaaa MULTI PATCH NUMBER 8
00 41H	0000 bbbb aaaa bbbb 0-31
:	
00 42H	0xxx xxxx dummy
:	
00 53H	0xxx xxxx
00 54H	0000 aaaa KEYBOARD DISPLAY

1: B
2: C
3: D
4: E
5: F
6: G
7: H
8: ALL

00 56H	0000 aaaa	MULTI LEVEL 1	
00 57H	0000 bbbb	aaaa bbbb	0-127
:	:	:	:
00 64H	0000 aaaa	MULTI LEVEL 8	
00 65H	0000 bbbb	aaaa bbbb	0-127
:	:	:	:
00 66H	0000 aaaa	BLOCK 1 DISK LABEL 1	
00 67H	0000 bbbb	aaaa bbbb	32-127 (ASCII)
:	:	:	:
01 5CH	0000 aaaa	BLOCK 1 DISK LABEL 60	
01 5DH	0000 bbbb	aaaa bbbb	32-127 (ASCII)
:	:	:	:
01 5EH	0xxx xxxx	dummy	
:	:	:	:
01 65H	0xxx xxxx		
:	:	:	:
01 66H	0000 aaaa	EXTERNAL CONTROLLER	
01 67H	0000 bbbb	aaaa bbbb	0: OFF 1: MOUSE 2: RC-100
:	:	:	:
01 68H	0xxx xxxx	dummy	
:	:	:	:
04 65H	0xxx xxxx		
:	:	:	:
04 66H	0000 aaaa	BLOCK 2 DISK LABEL 1	
04 67H	0000 bbbb	aaaa bbbb	32-127 (ASCII)
:	:	:	:
05 5CH	0000 aaaa	BLOCK 2 DISK LABEL 60	
05 5DH	0000 bbbb	aaaa bbbb	32-127 (ASCII)
:	:	:	:
05 5EH	0xxx xxxx	dummy	
:	:	:	:
07 7FH	0xxx xxxx		
Total size 00 00 08 00H			

4.3 MIDI parameter

Offset address	Description
00 00H	0xxx xxxx dummy
:	:
00 3FH	0xxx xxxx
:	:
00 40H	0000 aaaa RX CHANNEL 1
00 41H	0000 bbbb aaaa bbbb 0-16 0-15... 1-16 CH 16... OFF
:	:
00 4EH	0000 aaaa RX CHANNEL 8
00 4FH	0000 bbbb aaaa bbbb 0-16
:	:
00 50H	0000 aaaa RX PROGRAM CHANGE 1
00 51H	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
00 5EH	0000 aaaa RX PROGRAM CHANGE 8
00 5FH	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
00 60H	0000 aaaa RX BENDER 1
00 61H	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
00 6EH	0000 aaaa RX BENDER 8
00 6FH	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
00 70H	0000 aaaa RX MODULATION 1
00 71H	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
00 7EH	0000 aaaa RX MODULATION 8
00 7FH	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
01 00H	0000 aaaa RX HOLD 1
01 01H	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
01 0EH	0000 aaaa RX HOLD 8
01 0FH	0000 bbbb aaaa bbbb 0: OFF 1: ON
:	:
01 1EH	0000 aaaa RX AFTER TOUCH 1
01 1FH	0000 bbbb aaaa bbbb 0: OFF 1: ON

:	:	:	1: ON
01 1EH	0000 aaaa	RX AFTER TOUCH 8	
01 1FH	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 20H	0000 aaaa	RX VOLUME 1	
01 21H	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 2EH	0000 aaaa	RX VOLUME 8	
01 2FH	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 30H	0000 aaaa	RX BEND RANGE 1	
01 31H	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 3EH	0000 aaaa	RX BEND RANGE 8	
01 3FH	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 40H	0xxx xxxx	dummy	
01 41H	0xxx xxxx		
:	:	:	:
01 42H	0000 aaaa	SYSTEM EXCLUSIVE	
01 43H	0000 bbbb	aaaa bbbb	0: OFF 1: ON
:	:	:	:
01 44H	0000 aaaa	DEVICE ID	
01 45H	0000 bbbb	aaaa bbbb	0-15
:	:	:	:
01 46H	0000 aaaa	RX PROGRAM CHANGE NUMBER 1	
01 47H	0000 bbbb	aaaa bbbb	0-127
:	:	:	:
02 06H	0000 aaaa	RX PROGRAM CHANGE NUMBER 32	
02 07H	0000 bbbb	aaaa bbbb	0-127
:	:	:	:
02 08H	0xxx xxxx	dummy	
:	:	:	:
03 7FH	0xxx xxxx		
Total size 00 00 04 00H			

4.4 Tone parameters

Offset address	Description
00 00H	0000 aaaa TONE NAME 1
00 01H	0000 bbbb aaaa bbbb 32-127 (ASCII)
:	:
00 0EH	0000 aaaa TONE NAME 8
00 0FH	0000 bbbb aaaa bbbb 32-127 (ASCII)
:	:
00 10H	0000 aaaa OUTPUT ASSIGN
00 11H	0000 bbbb aaaa bbbb 0-7
:	:
00 12H	0000 aaaa SOURCE TONE
00 13H	0000 bbbb aaaa bbbb 0-31
:	:
00 14H	0000 aaaa ORIG/SUB TONE
00 15H	0000 bbbb aaaa bbbb 0: ORG 1: SUB
:	:
00 16H	0000 aaaa SAMPLING FREQUENCY
00 17H	0000 bbbb aaaa bbbb 0: 30kHz 1: 15kHz
:	:
00 18H	0000 aaaa ORIG KEY NUMBER
00 19H	0000 bbbb aaaa bbbb 11-120 (MIDI FORMAT)
:	:
00 1AH	0000 aaaa WAVE BANK
00 1BH	0000 bbbb aaaa bbbb 0: A 1: B
:	:
00 1CH	0000 aaaa WAVE SEGMENT TOP
00 1DH	0000 bbbb aaaa bbbb 0-17
:	:
00 1EH	0000 aaaa WAVE SEGMENT LENGTH
00 1FH	0000 bbbb aaaa bbbb 0-18
:	:
00 20H	0000 aaaa START POINT
00 21H	0000 bbbb aaaa bbbb
00 22H	0000 cccc aaaa bbbb cccc dddd eeee ffff
00 23H	0000 dddd
00 24H	0000 eeee
00 25H	0000 ffff
:	:
00 26H	0000 aaaa END POINT
00 27H	0000 bbbb aaaa bbbb
00 28H	0000 cccc aaaa bbbb cccc dddd eeee ffff
00 29H	0000 dddd
00 2AH	0000 eeee
00 2BH	0000 ffff

01 2EH	0000 aaaa	COPY SOURCE	
01 2FH	0000 bbbb	aaaa bbbb	0 - 31
01 30H	0000 aaaa	LOOP TUNE	
01 31H	0000 bbbb	aaaa bbbb	- 64 - + 63
01 32H	0000 aaaa	TVA LEVEL CURVE	
01 33H	0000 bbbb	aaaa bbbb	0 - 5
01 34H	0xxx xxxx	dummy	
:			
01 4BH	0xxx xxxx		
01 4CH	0000 aaaa	LOOP LENGTH	
01 4DH	0000 bbbb		
01 4EH	0000 cccc	aaaa bbbb cccc dddd eeee ffff	
01 4FH	0000 dddd		000004 - 221184
01 50H	0000 eeee		
01 51H	0000 ffff		
01 52H	0000 aaaa	PITCH FOLLOW	
01 53H	0000 bbbb	aaaa bbbb	0 : OFF 1 : ON
01 54H	0000 aaaa	ENV ZOOM	
01 55H	0000 bbbb	aaaa bbbb	0 - 5
01 56H	0000 aaaa	TVF ENV SUSTAIN POINT	
01 57H	0000 bbbb	aaaa bbbb	0 - 7
01 58H	0000 aaaa	TVF ENV END POINT	
01 59H	0000 bbbb	aaaa bbbb	1 - 7
01 5AH	0000 aaaa	TVF ENV LEVEL 1	
01 5BH	0000 bbbb	aaaa bbbb	0 - 127
01 5CH	0000 aaaa	TVF ENV RATE 1	
01 5DH	0000 bbbb	aaaa bbbb	1 - 127
01 5EH	0000 aaaa	TVF ENV LEVEL 2	
01 5FH	0000 bbbb	aaaa bbbb	0 - 127
01 60H	0000 aaaa	TVF ENV RATE 2	
01 61H	0000 bbbb	aaaa bbbb	1 - 127
01 62H	0000 aaaa	TVF ENV LEVEL 3	
01 63H	0000 bbbb	aaaa bbbb	0 - 127
01 64H	0000 aaaa	TVF ENV RATE 3	
01 65H	0000 bbbb	aaaa bbbb	1 - 127
01 66H	0000 aaaa	TVF ENV LEVEL 4	
01 67H	0000 bbbb	aaaa bbbb	0 - 127
01 68H	0000 aaaa	TVF ENV RATE 4	
01 69H	0000 bbbb	aaaa bbbb	1 - 127
01 6AH	0000 aaaa	TVF ENV LEVEL 5	
01 6BH	0000 bbbb	aaaa bbbb	0 - 127
01 6CH	0000 aaaa	TVF ENV RATE 5	
01 6DH	0000 bbbb	aaaa bbbb	1 - 127
01 6EH	0000 aaaa	TVF ENV LEVEL 6	
01 6FH	0000 bbbb	aaaa bbbb	0 - 127
01 70H	0000 aaaa	TVF ENV RATE 6	
01 71H	0000 bbbb	aaaa bbbb	1 - 127
01 72H	0000 aaaa	TVF ENV LEVEL 7	
01 73H	0000 bbbb	aaaa bbbb	0 - 127
01 74H	0000 aaaa	TVF ENV RATE 7	
01 75H	0000 bbbb	aaaa bbbb	1 - 127
01 76H	0000 aaaa	TVF ENV LEVEL 8	
01 77H	0000 bbbb	aaaa bbbb	0 - 127
01 78H	0000 aaaa	TVF ENV RATE 8	
01 79H	0000 bbbb	aaaa bbbb	1 - 127
01 7AH	0000 aaaa	AFTER TOUCH SWITCH	
01 7BH	0000 bbbb	aaaa bbbb	0 : OFF 1 : ON
01 7CH	0xxx xxxx	dummy	
:			
01 7FH	0xxx xxxx		
Total size		00 00 02 00H	

4.5 SW

Offset address	Description
00H	0000 aaaa SW 1 (all)
01H	0000 bbbb aaaa bbbb
02H	0000 aaaa SW 2 (character)
03H	0000 bbbb aaaa bbbb
04H	0000 aaaa SW 3 (patch)
05H	0000 bbbb aaaa bbbb
06H	0000 aaaa ALPHA DIAL
07H	0000 bbbb aaaa bbbb - 127 - + 127
Total size	00 00 00 08H

4.6 Wave data

Offset address	Description
00 00 00H	0aaa aaaa aaaa aaab bbbb
00 00 01H	0bbb bb00 12 bit 2's complement data
:	
:	
01 3F 7EH	
01 3F 7FH	
Total size	00 01 40 00H